[CSL252]-Design and Analysis of Algorithms **2025-26-W**

TUTORIAL X

Date of the tutorial: **Apr 02, 2026**.

Due Date: **11:59 pm, Apr 06, 2026**.

Instructions. Same as Tutorial I

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1. Define the following terms. (a) Vertex cover (b) Independent set (c) Clique. What is the relationship (if any) between them. Justify your claim.
2. Define the following terms (a) P (b) NP (c) NP-complete and (d) NP-hard.
3. Show that the following problems are in NP.
   (a) VERTEX COVER
   (b) INDEPENDENT SET
   (c) CLIQUE
   (d) HAMILTONIAN PATH
   (e) k-coloring, ($k \geq 3$)
   (f) 3-SAT
4. Formulate a decision version for the CLIQUE problem, and prove that it is NP-complete. (give a reduction from 3-SAT)
5. (a) Describe a polynomial-time reduction from the HAMILTONIAN PATH problem to HAMILTONIAN CYCLE.
   (b) Describe a polynomial-time reduction from the HAMILTONIAN CYCLE problem to HAMILTONIAN PATH
6. Formulate both optimization and decision version of TRAVELLING SALESMAN problem. Show that the decision problem is NP-complete. (give a reduction from Hamiltonian cycle)
7. Let $G = (V, E)$ be a graph with a weight function $w : V \to \mathbb{R}^+$.
   A set $X \subseteq V$ is a vertex cover if for every edge $(u, v) \in E$, $u \in C$ or $v \in C$.
   The WEIGHTED VERTEX COVER problem is to find a vertex cover $C \subseteq V$ that minimizes$\sum_{v \in C} w(v)$.
   A set $I \subseteq V$ is an *independent set* if for every edge $(u, v) \in E$,
   $\{u, v\} \nsubseteq I$ (i.e., not both endpoints are in I).
   The WEIGHTED INDEPENDENT SET problem is to find an independent set $I \subseteq V$ that maximizes $\sum_{v \in I} w(v)$.
   Give a polynomial time algorithm to solve the (a) WEIGHTED INDEPENDENT SET and (b) WEIGHTED VERTEX COVER problems when G is weighted tree.